

Data Flow Diagram

Date	15 July 2026
Team ID	SWTID-2026-2749
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

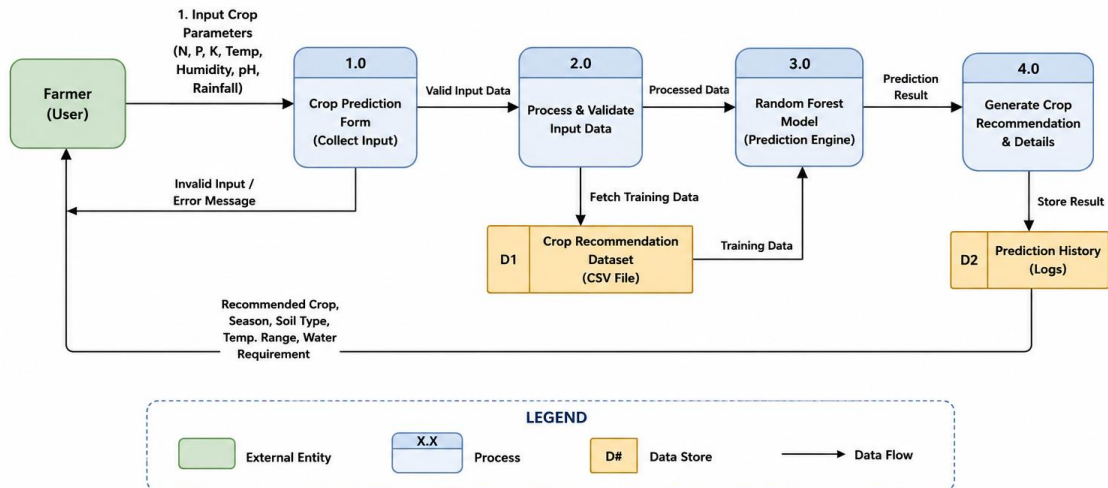
Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

Data Flow Diagram (DFD) – Level 0

OptiCrop: Smart Agricultural Production Optimization Engine



Description of Flow:

1. The farmer enters soil and weather parameters into the system.
2. The system validates the input data.
3. Valid data is processed by the Random Forest model which uses the trained dataset.
4. The system generates the recommended crop along with details and shows the result to the farmer.
5. The prediction result is stored in the prediction history for future reference.